

L272 IS AUTO COIL MACHINE

OPERATION MANUAL

MACHINE FRONT PANEL

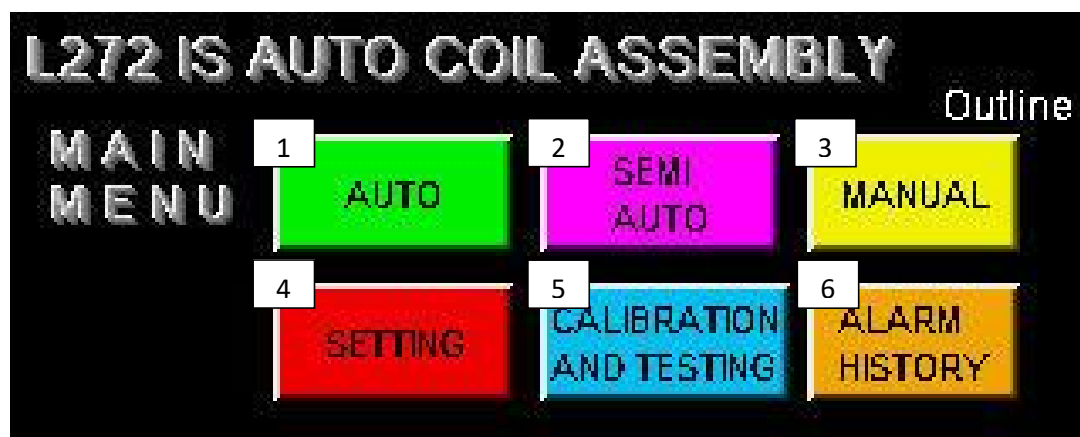


Figure 1 : Main panel

MAIN PANEL

No	Function	Description
1	AUTO MODE	WHEN TURNED ON, THE MACHINE IS IN AUTO MODE AND READY FOR PRODUCTION. WHEN TURNED OFF, THE MACHINE IS IN DEBUGGING OR REPAIRING MODE.
2	SEMI AUTO MODE	TO ENTER SEMI AUTO MODE
3	MANUAL	TO ENTER IO & SERVO SETTING MODE
4	SETTING	TO ENTER SETTING MODE
5	CALIBRATION AND TESTING	TO ENTER DRIVER CALIBRATION MODE
6	ALARM HISTORY	TO CHECK ALARM HISTORY

MACHINE OPERATION PANEL

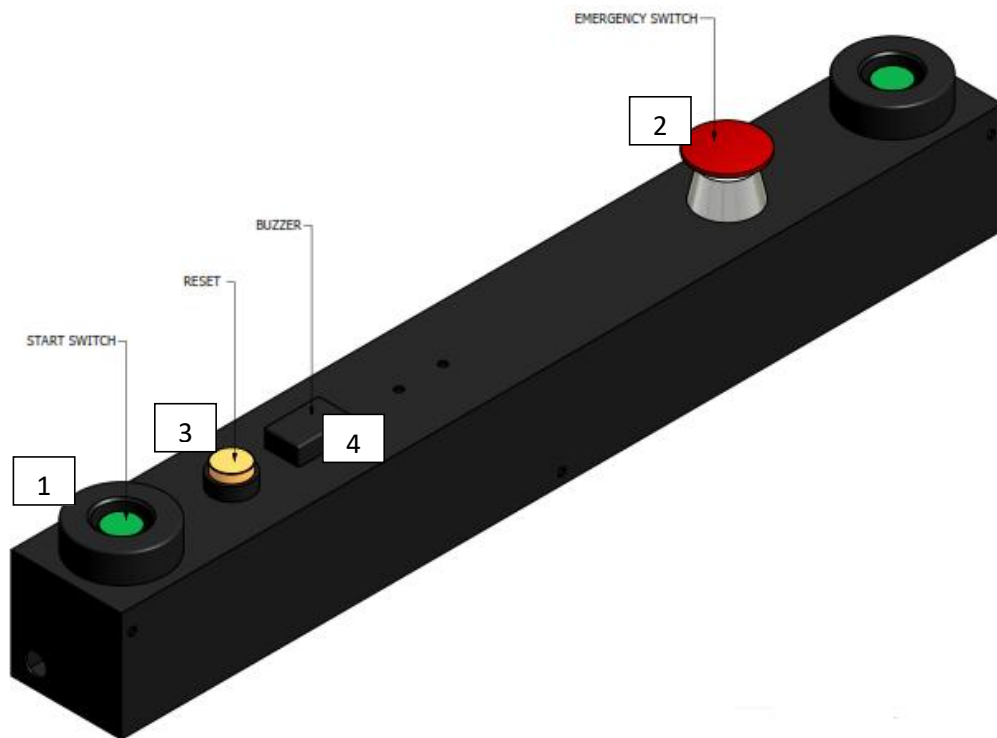


Figure 2 : Operation panel

OPERATION PANEL

No	Function	Description
1	START BUTTON	PRESS TO START THE MACHINE
2	EMERGENCY BUTTON	PRESS TO EMERGENCY STOP
3	RESET BUTTON	PRESS TO RESET THE MACHINE AND BUZZER
4	BUZZER	WILL TURN ON WHEN PROCESS COMPLETE OR EMERGENCY STOP

MACHINE OPERATION PROCEDURE

1. POWER ON THE MACHINE BY SWITCH ON POWER BUTTON.
2. TURN ON AIR SUPPLY TO THE MACHINE AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.
3. PRESS RESET BUTTON AND MACHINE WILL PERFORM HOMING ROUTINE.
4. GOTO SETTING MODE AND SELECT SLIDER SPEED.
5. SET MACHINE TO AUTO MODE.
6. LOAD PRODUCT IN MACHINE AND PRESS THE START BUTTON.
7. WHEN PROCESS IS COMPLETED, THE BUZZER WILL SOUND 3 TIMES AND OPERATOR CAN UNLOAD THE PRODUCT FROM THE MACHINE.
8. REPEAT PROCESS 5 AND 6 TO CONTINUE PRODUCTION.

SETTING FUNCTION



Figure 3 : Speed Screen

SPEED SETTING SCREEN

No	Buttons	Description
1	120%	Machine run at 120% of normal speed
2	100%	Machine run at 10% of normal speed
3	80%	Machine run at 80% of normal speed
4	BACK	Back to previous page
5	NEXT	Go to next page



10

Figure 4 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	LOAD	To start the tray load sequence
2	UNLOAD	To start the tray unload sequence
3	GLUE	To start the glue sequence
4	UV	To start the UV sequence
5	BASE	To start the base load sequence at selected no
6	COIL 1 PICK & COIL 1 PLACE	To start the coil 1 load sequence at selected no
7	COIL 2 PICK & COIL 2 PLACE	To start the coil 2 load sequence at selected no



Figure 5 : Timing Screen

TIMING SCREEN

No	Function	Description
1	UV ON	Timing for UV ON
2	PICK TIMEOUT	Timeout for coil pick
3	PART TIMEOUT	Timeout for base/part pick
4	GLUE	Timing for GLUE
5	SLOW PULSE	Speed change for Z Axis point
6	RETRY	Retry for coil pick



Figure 6 : Servo Speed Screen

SERVO SPEED SCREEN

No	Function	Description
1	ACC/DEC	Acc/Decc speed for each of the axis
2	OPERATING	Operating speed for each of the axis

SEMI AUTO FUNCTION



Figure 7 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	BASE	Insert Base process will be turned on when selected
2	COIL 1 PICK	Pick Coil 1 process will be turned on when selected
3	COIL 1 PLACE	Place Coil 1 process will be turned on when selected
4	COIL 2 PICK	Pick Coil 1 process will be turned on when selected
5	COIL 2 PLACE	Place Coil 2 process will be turned on when selected
6	GLUE	Apply Glue process will be turned on when selected
7	UV	UV process will be turned on when selected

MANUAL FUNCTION



Figure 8 : Semi Auto Screen

MANUAL SCREEN

No	Function	Description
1	I/O CONTROL	Go to I/O Control screen
2	SERVO CONTROL	Go to Servo Control screen

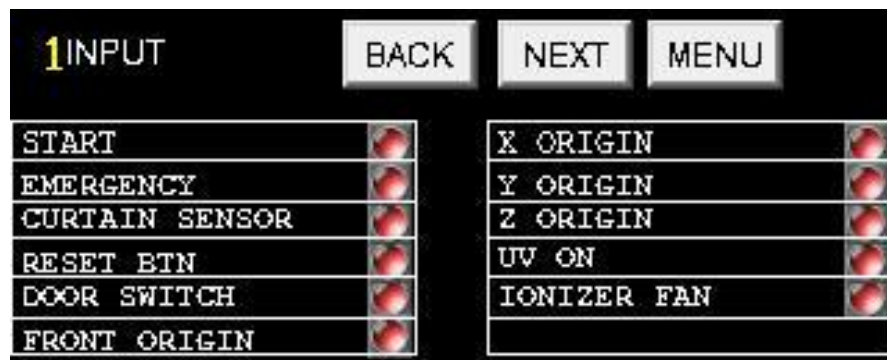


Figure 9 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals



Figure 10 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals



Figure 11 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals



Figure 12 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals



Figure 13 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

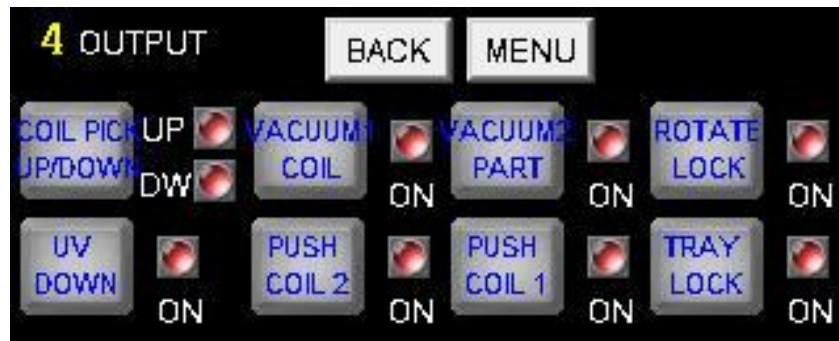


Figure 14 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals



Figure 15 : Servo Control Menu Screen

SERVO CONTROL SCREEN

No	Function	Description
1	TOP TRAY POSITION	Go to screen to teach coil pick up position
2	BTM TRAY POSITION	Go to screen to teach base pick up position
3	ROTATE POSITION	Go to screen to teach tray drop position
4	PART SLIDE	Go to screen to teach part carrier slide position
5	TOP TRAY SLIDE	Go to screen to teach coil tray position
6	TOP TRAY PITCH	Go to screen to setup coil tray dimensions & servo calculation specs
7	BTM TRAY PITCH	Go to screen to setup base tray dimensions & servo calculation specs

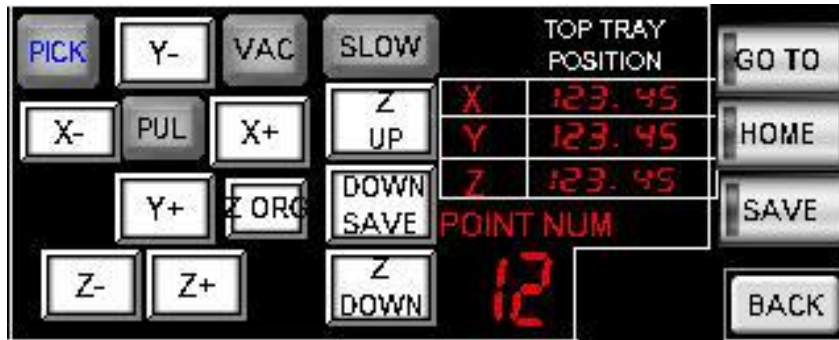


Figure 16 : Top Tray Position Screen

TOP TRAY SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse
13	VAC	Turn on vacuum
14	GOTO	Move X & Y Axis to selected point no.
15	HOME	Move X,Y & Z Axis to Origin position
16	SAVE	Save X-Axis & Y-Axis to selected point no.

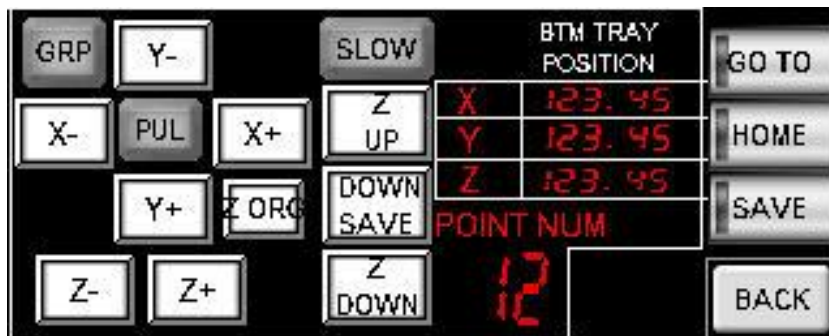


Figure 17 : Btm Tray Position Screen

BTM TRAY SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	GRP	Turn on pick gripper
12	PUL	Turn on pulse
13	GOTO	Move X & Y Axis to selected point no.
14	HOME	Move X,Y & Z Axis to Origin position
15	SAVE	Save X-Axis & Y-Axis to selected point no.



Figure 18 :Rotate Position Screen

ROTATE TRAY SCREEN

No	Function	Description
1	UP	Move Rotate Axis Up
2	DW	Move Rotate Axis Down
3	SLOW	Make jog slow or fast
4	TRAY LOCK	Turn on top tray lock
5	TRAY SUPPORT	Turn on bottom tray support
6	GOTO	Move Rotate Axis to selected point no.
7	HOME	Move Rotate Axis to Origin position
8	SAVE	Save Rotate Axis to selected point no.



Figure 19 :Rotate Position Screen

PART CARRIER SLIDE SCREEN

No	Function	Description
1	X+	Move Part Carrier Slide to the right
2	X-	Move Part Carrier Slide to the left
3	i	Show information of all the point no.
4	GOTO	Move Part Carrier Axis to selected point no.
5	HOME	Move Part Carrier Axis to Origin position
6	SAVE	Save Part Carrier Axis to selected point no.



Figure 20 :Top Tray Slide Screen

TOP TRAY SLIDE SCREEN

No	Function	Description
1	GOTO	Move Top Tray Slide Axis to selected point no.
2	HOME	Move Top Tray Slide Axis to Origin position
3	SAVE	Save Top Tray Slide Axis to selected point no.



Figure 21 : Top Tray Pitch Setting Screen

TOP TRAY PICTH SCREEN

No	Function	Description
1	X PITCH	Top Tray(Coil Tray) X pitch
2	Y PITCH	Top Tray(Coil Tray) Y pitch
3	ROW	The number of coil rows
4	COL	The number of coil columns
5	TOTAL	Total number of coils(auto calculated)
6	1mm(X)	Servo X pulse specs
7	1mm(Y)	Servo Y pulse specs



Figure 22 : Bottom Tray Pitch Setting Screen

BOTTOM TRAY PICTH SCREEN

No	Function	Description
1	X PITCH	Bottom Tray(Base Tray) X pitch
2	Y PITCH	Bottom Tray(Base Tray) Y pitch
3	ROW	The number of base rows
4	COL	The number of base columns
5	TOTAL	Total number of base (auto calculated)
6	1mm(X)	Servo X pulse specs
7	1mm(Y)	Servo Y pulse specs

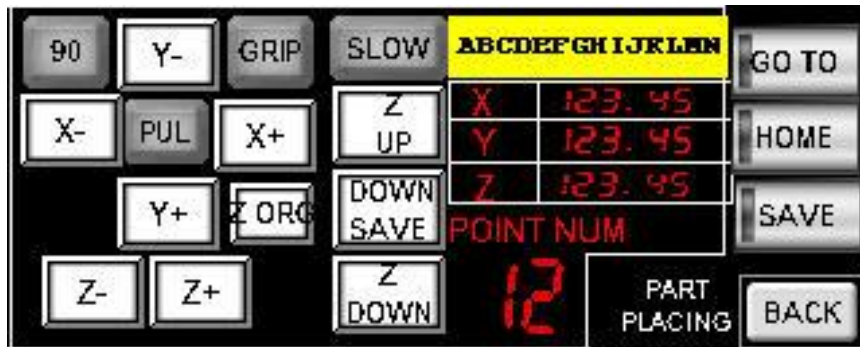


Figure 23 : PartPlacing Screen

PART PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	GRIP	Turn on pick gripper
12	90	Turn on rotate cylinder
13	GOTO	Move X & Y Axis to selected point no.
14	HOME	Move X,Y & Z Axis to Origin position
15	SAVE	Save X-Axis & Y-Axis to selected point no.

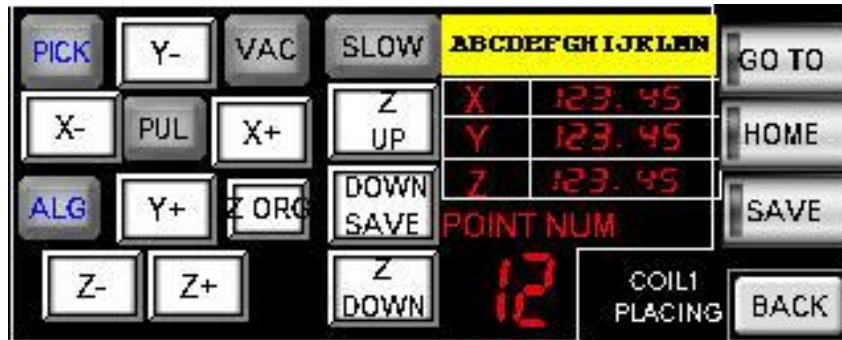


Figure 24 : Coil 1 Placing Screen

COIL 1 PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse
13	VAC	Turn on vacuum
14	GOTO	Move X & Y Axis to selected point no.
15	HOME	Move X,Y & Z Axis to Origin position
16	SAVE	Save X-Axis & Y-Axis to selected point no.

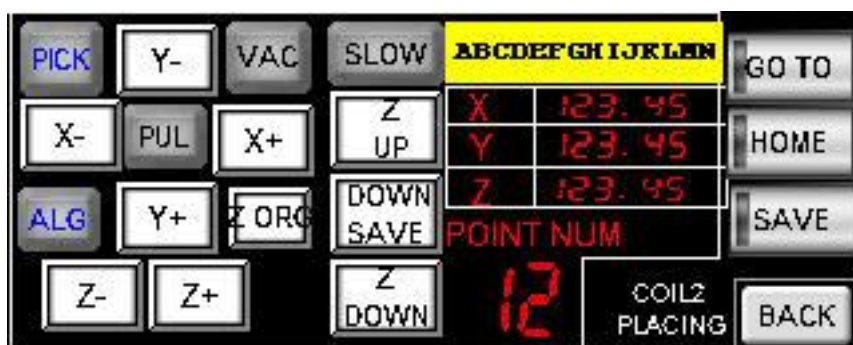


Figure 24 : Coil 2 Placing Screen

COIL 2 PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse
13	VAC	Turn on vacuum
14	GOTO	Move X & Y Axis to selected point no.
15	HOME	Move X,Y & Z Axis to Origin position
16	SAVE	Save X-Axis & Y-Axis to selected point no.

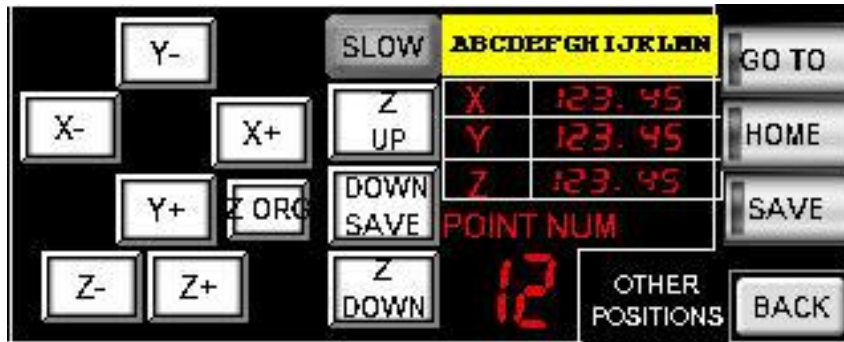


Figure 25 : Other Position Placing Screen

OTHER POSITION PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	GOTO	Move X & Y Axis to selected point no.
12	HOME	Move X,Y & Z Axis to Origin position
13	SAVE	Save X-Axis & Y-Axis to selected point no.

MACHINE PLC SETTING & LADDER DIAGRAM

0	1	2	3	4	5	End Unit	
KV-S000	KV-L2*V	KV-B16X+	KV-B16X+	KV-B16R+	KV-B16R+		
R000 -015		R33000 -33015	R34000 -34015				
R500 -507				R35000 -35015	R36000 -36015		
R30000 -31915	R32000 -32115						

Unit

Select unit(1)

Setup unit(2)

[1] KV-L2*V

Base

Leading DM No.

DM10100

Number of DMs in use

1

Leading relay No.(set channel ...

R32000

Number of relays in use

32

Node No.

0(*)

Port 1

Operation mode

KV BUILDER/KV STUDIO mode

Detail

---(*)

Interface

RS-232C(*)

Baudrate

Auto

Data bit length

8 bits(*)

Start bit

1 bits(*)

Stop bit

1 bits(*)

Parity

Even(*)

Checksum

none(*)

RS/CS flow control

OFF(*)

Port 2

Operation mode

KV BUILDER/KV STUDIO mode

Detail

---(*)

Interface

RS-232C

Baudrate

Auto

Data bit length

8 bits(*)

Start bit

1 bits(*)

Stop bit

1 bits(*)

Parity

Even(*)

Checksum

none(*)

Details setting

Transfer timeout time (secs)

3

Leading DM No.

To set up the leading No. of data memories (DM) to be used in this unit.

0	1	2	3	4	5	End Unit	
KV-5000	KV-L2*V	KV-B16X*	KV-B16X*	KV-B16R*	KV-B16R*		
R000 -015		R33000 -33015	R34000 -34015				
R500 -507				R35000 -35015	R36000 -36015		
R30000 -31915	R32000 -32115						

Unit 4

Select unit(1)

Setup unit(2)

[0] KV-5000

Function

Communication mode	Ethernet (*)
Extended function	Not used (*)
Mail setting	<setting>
Simple PLC link setting	<Setting>
FTP client setting	<Setting>

Unit

Select unit(1)
Setup unit(2)

[0] KV-5000

Base

Leading DM No.	DM10000
Number of DMs used	16
Leading relay No. (ch unit set...	R30000
Number of relays used	320
Baud rate	100/10 Mbps, Auto(*)
Setting method of IP address	Fixed IP address(*)
IP address	192.168.0.10
Subnet mask	255.255.255.0
Default gateway	0.0.0.0
Port No. (KVS, KV COM+, DB)	8500
Port No. (host link)	8501
Reception timeout[s].	10
Keep Alive[s]	600
FTP server enabled	disabled(*)
Password	[Unset]
FTP server RUN/PROG switching	Yes(*)
Routing setting	no control(*)
Port No. (VT)	8502
Port No. (System expansion)	8504
DNS Server	0.0.0.0
MC protocol port No. (TCP)	5000
MC protocol Port No. (UDP)	5000
MC protocol code	BIN(*)
Type name code of MC protocol	0
Type name of MC protocol	
Simple PLC link port No. (UDP)	0
Auto adjust clock	Not used(*)
SNIP communication timeout[ms]	60
Clock adjustment time time[h]	0
Clock adjustment time[m]	0
Clock adjustment time[s]	0
Clock adjustment interval[m]	60
GMT offset	+(*)
GMT offset[h]	9
GMT offset[m]	0
GMT offset[s]	0
NTP(SNTP) server	0.0.0.0

Socket1

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket2

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket3

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket4

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket5	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
Socket6	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
Socket7	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
Common KV socket	
Response timeout[s]	30

Introduction

This section describes the features and specifications of the Is Auto Coil Machine.

About IS Auto Coil Machine

The purpose of the system is to assembly , the base with 2 pcs of solenoid coil and apply glue with bonding.

- Simple operation

The system is designed and emphasizes on simple, easy teaching and automatic operations.

- Modern and user-friendly user interface

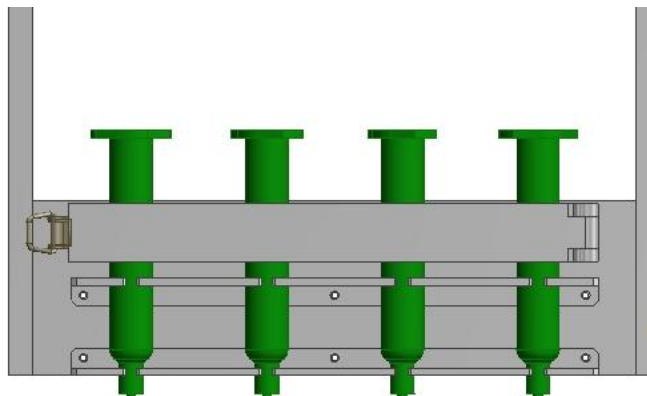
Simple and modern GUI of the controlling software features many functions and even easy to use.

System Specifications

The system specification will be listed below

Mechanical and Electrical Specifications

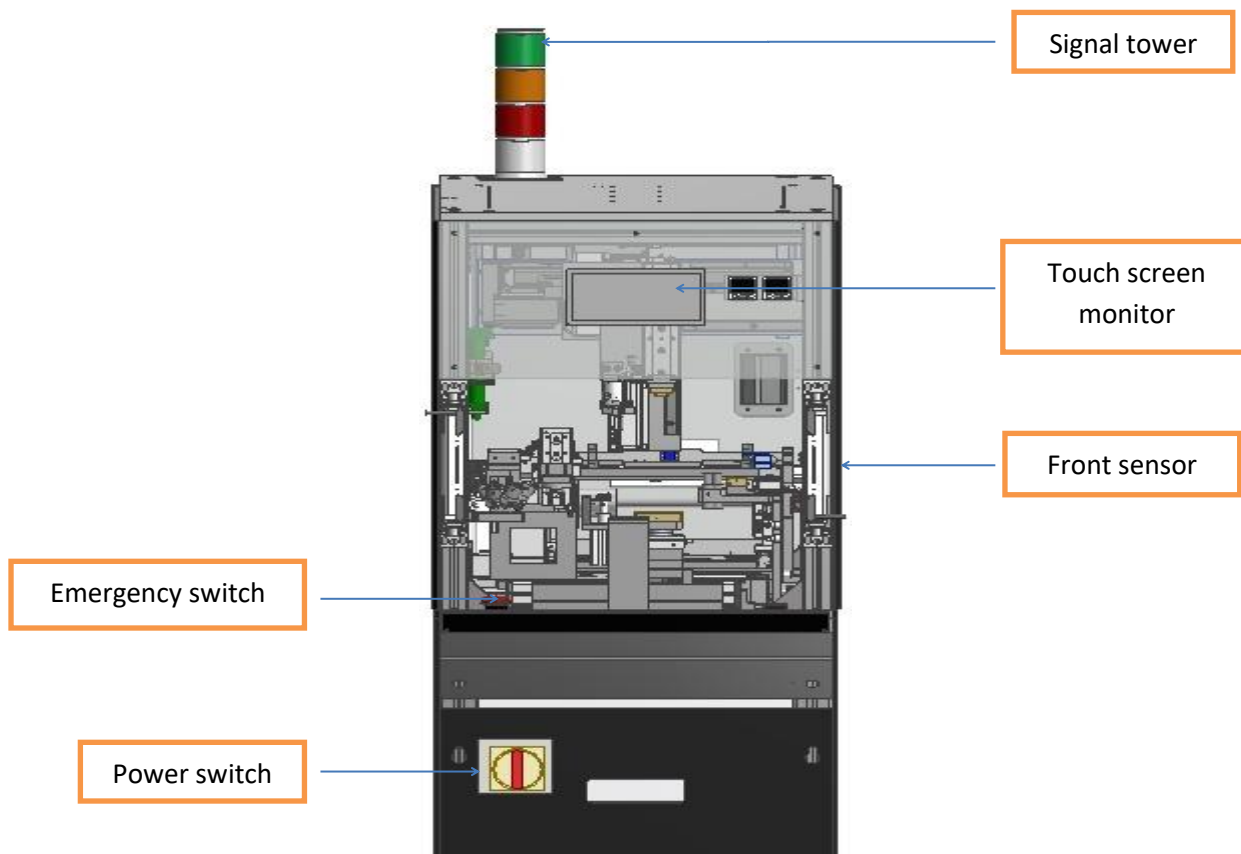
Item	Specifications
Control PLC	Keyence PLC
Power supply	100V AV - 110V AC
Air pressure	Min 0.5 Mpa or 5 bar (dry air)
Cycle time/pcs	49s/pcs



Names and functions of components

This section main describes names and the functions of the various parts of Is Auto Coil Machine.

External View



- Emergency switch

By pressing the emergency switch will bring the machine to emergency halt status.

- Monitor and touch screen

The touch screen monitor is used for machine operations control.

- Front curtain sensor

It is a safety prevention tool, only enabled when performing sorting operation. The operation will be halted if the sensor is **ON**.

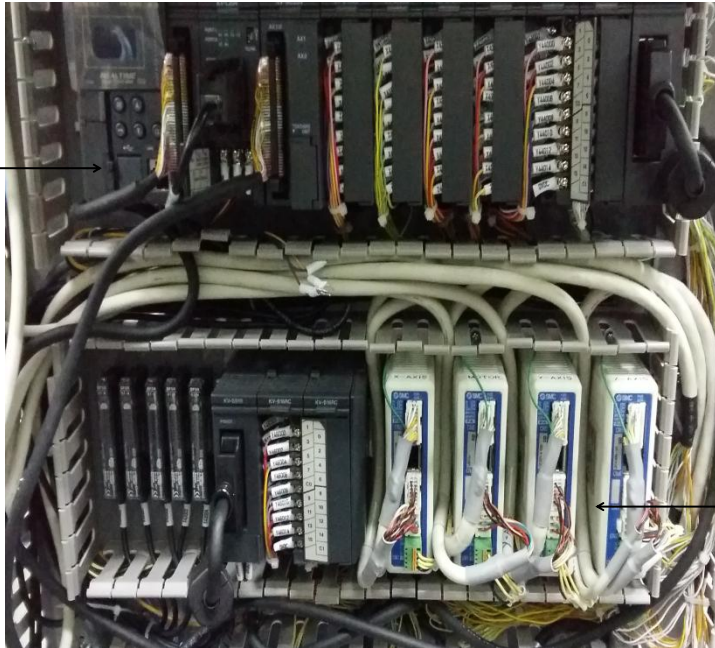
- Signal tower

It shows the system current status with the green, red and yellow lights with buzzer.

Inside Door

Power Box

Keyence PLC
KV 5000

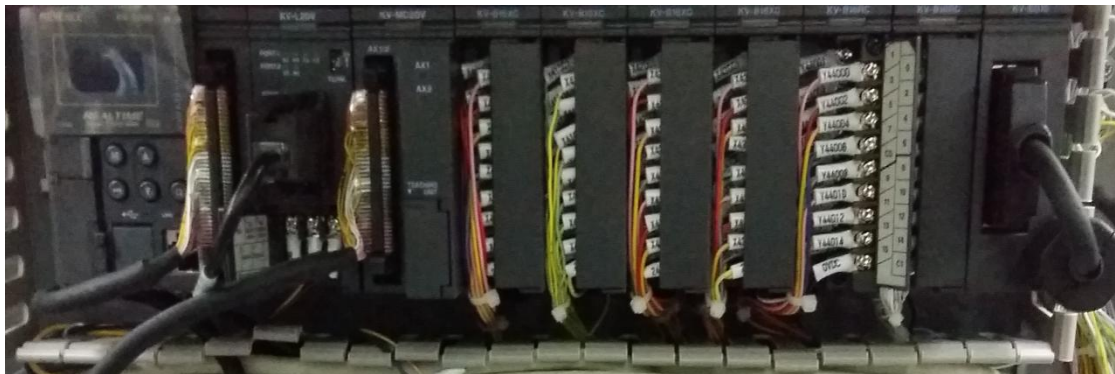


SMC Driver Motor
LE

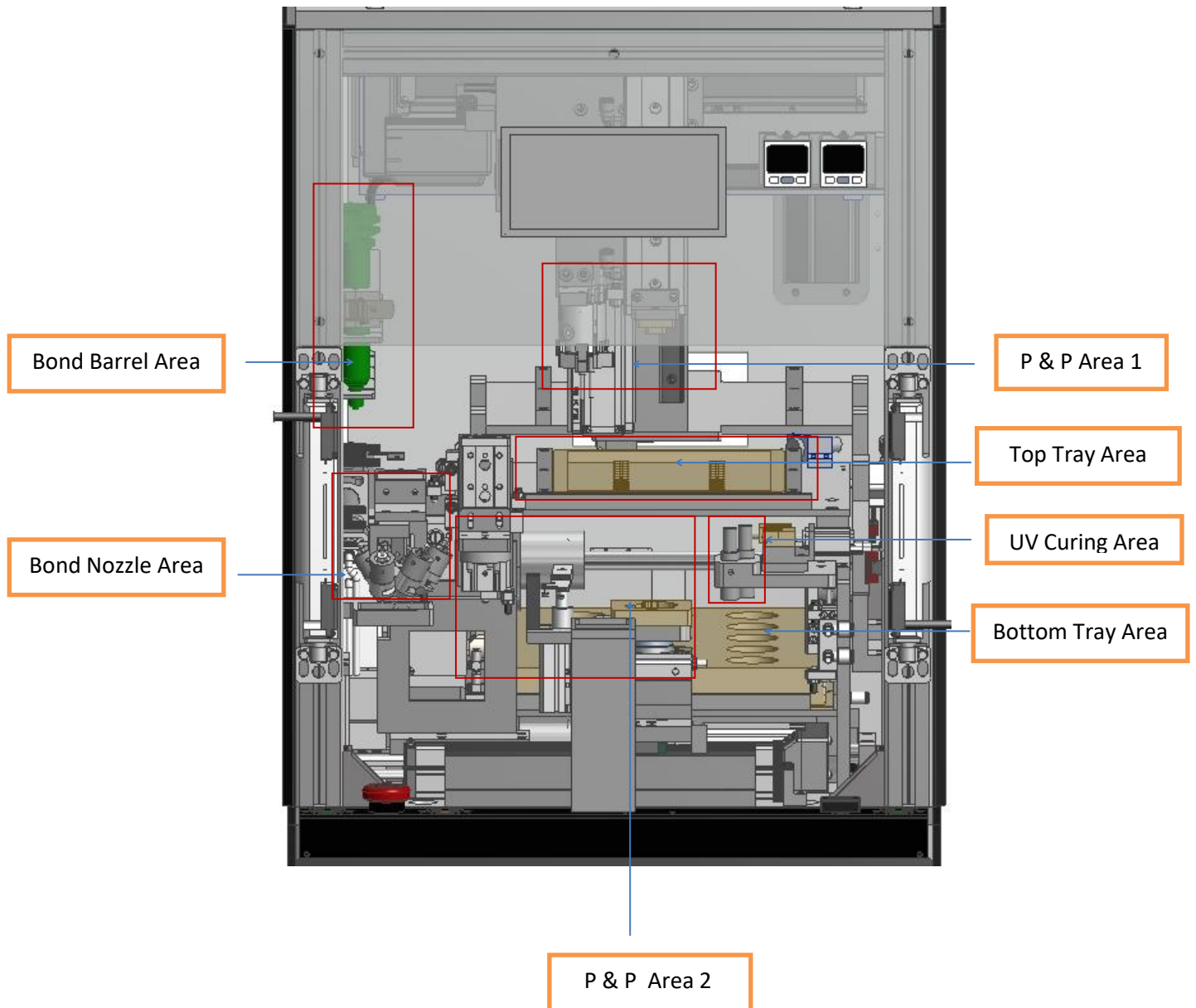
- SMC Servo Motor LE



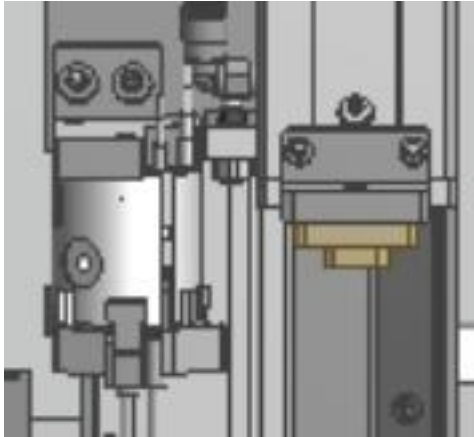
- Keyence PLC KV 5000



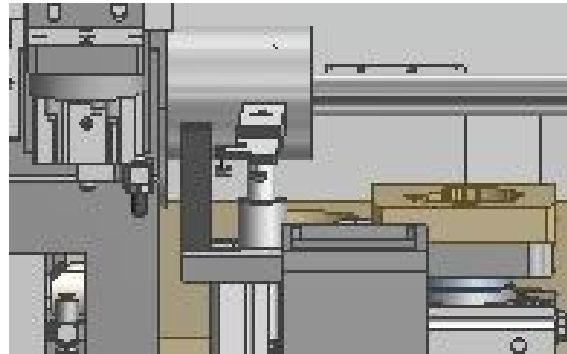
Working Section



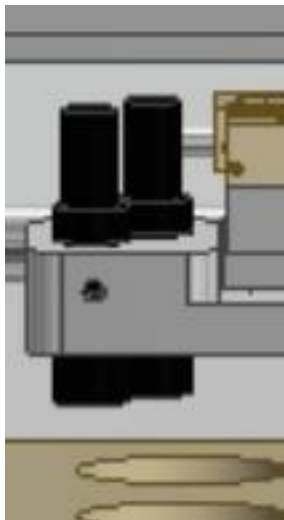
P & P Area 1



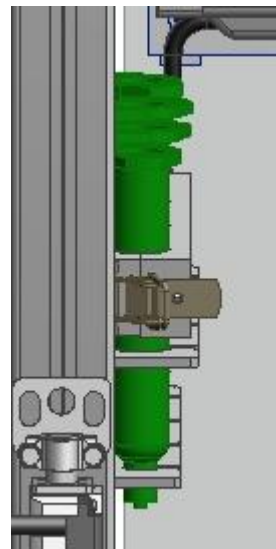
P & P Area 2



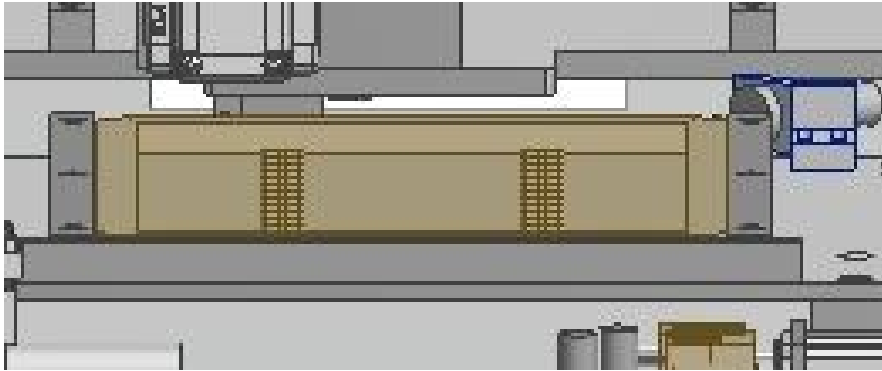
UV Curing Area



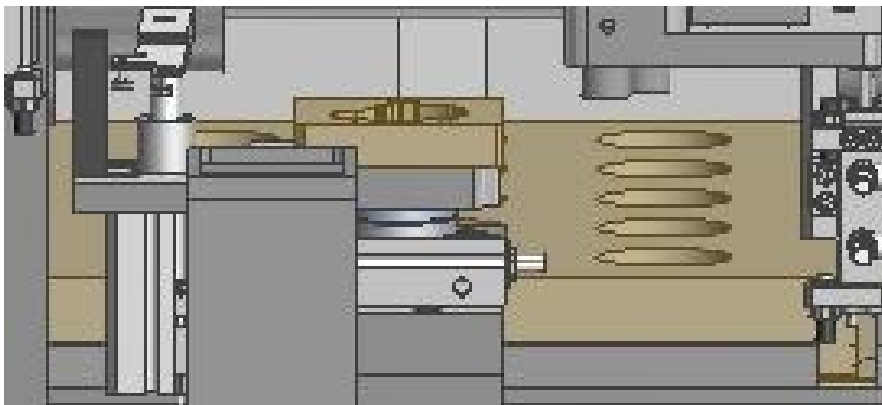
Bond Barrel Area



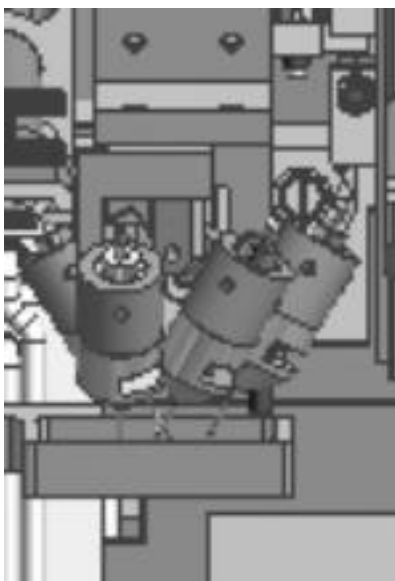
Top Tray Area



Bottom Tray Area



Bond Nozzle Area



OPERATION MANUAL

Table of contents

ITEM	LIST
1	Introduction
2	Machine Operation Panel
3	Touch Screen Panel
4	Wiring Diagram
5	Part List

INTRODUCTION

TOUCH SCREEN PANEL

WIRING DIAGRAM

PART LIST

L272 IS AUTO COIL MACHINE



OPERATION MANUAL

MACHINE OPERATION PANEL